

AIGR Gene Hypothesis Deep Research — LPGAT1 (Q92604)

Hypothesis slug: secondary-acylation-capacities-and-lipid-pathway-scope **Focus type:** function_assignment **Gene:** LPGAT1 / LPLAT7 (Homo sapiens, NCBITaxon:9606), UniProt Q92604

Executive Judgment

Verdict: Partially supported.

Human LPGAT1's demonstrable, reproducible primary molecular function is that of an **endoplasmic-reticulum, sn-1-specific, stearoyl-CoA-preferring lysophospholipid acyltransferase acting on lysophosphatidylethanolamine (LPE) and lysophosphatidylcholine (LPC)**. This is established by three independent groups using different constructs, organisms, and readouts (recombinant human LPLAT7 screen, bacterially expressed murine LPGAT1, and skeletal-muscle knockout lipidomics). Against this core, the seed hypothesis proposes three "secondary" capacities: (1) monoacylglycerol acyltransferase (MGAT)/triacylglycerol (TAG) biosynthesis, (2) PG/PI remodeling, and (3) acylation of 1-acyl-lysophosphatidylserine (LPS). Assessed separately with attention to donor/acceptor and sn-position, these secondary activities are **real-but-re-ranked, context-specific, in-vivo-only, or paralog-confounded — not co-equal primary functions**.

The seed's central question — *"does newer predominant positional specificity exclude all older secondary activities, or do construct/assay/organism differences explain them?"* — is answered **the latter, with important nuance**. The newer sn-1 LPE/LPC specificity **re-ranks** rather than **erases** the older activities. The phrase "poor is not zero" from

P 36049524

is correct and matters for curation: LPI/LPG are weak but detectable intrinsic substrates, and the same paper's experimental evidence underlies the curated LPS reactions. However, several older annotations rest on weaker foundations than the sn-1 LPE/LPC core: MGAT is by-

similarity to mouse (single human study), TAG "biosynthesis" is a downstream pathway-flux phenotype (knockout *raises* TAG), and PI/LPI remodeling has no curated UniProt reaction and is better attributed to the same-family paralog ALCAT1/LCLAT1.

Most important caveats: (i) sn-position assignments for lyso-substrates are confounded by non-enzymatic acyl migration, so "sn-1 vs sn-2" claims must be read cautiously; (ii) the seed's exact phrase "acylates 1-acyl-lysophosphatidylserine" maps to a curated RHEA reaction (RHEA:37403) but with weak experimental backing; (iii) in-vivo lipidomic phenotypes reflect pathway flux and abundance, not necessarily direct intrinsic chemistry, and must not be curated as MF evidence.

Key Findings

Finding 1 — The core function is an ER sn-1-specific LPE/LPC acyltransferase, stearoyl-CoA preferring

Three independent lines of evidence converge on a single, tightly specified molecular function.

In vitro recombinant human screen (PMID:36049524). In a systematic siRNA + recombinant screen of 14 human LPLATs, LPLAT7/LPGAT1 was identified as an sn-1-specific lysophospholipid acyltransferase. The verified snippet is unambiguous: LPLAT7 *"mainly incorporated several fatty acids into the sn-1 position of lysophosphatidylcholine (LPC) and lysophosphatidylethanolamine (LPE), with weak activities toward other lyso-PLs."* In cells and knockout tissue, only C18:0 (stearoyl) PC/PE species were reduced, with a reciprocal rise in C16:0/C18:1 species — a signature of loss of a stearate-preferring sn-1 remodeling enzyme.

Independent recombinant murine assay (PMID:35131264). Bacterially expressed murine LPGAT1 *"transferred saturated acyl-CoAs specifically into the sn-1 position of lysophosphatidylethanolamine (LPE) rather than lysophosphatidylglycerol and preferred stearoyl-CoA over palmitoyl-CoA as the substrate."* Knockout abolished 1-LPE:stearoyl-CoA activity and shifted PE/PC composition from stearate toward palmitate. This directly demonstrates sn-1 LPE specificity, stearoyl-CoA preference, and — critically for the PG hypothesis — activity into LPE *rather than* LPG.

In vivo/ex vivo confirmation (PMID:37217003). In murine skeletal muscle, LPGAT1 sets the sn-1 18:0/16:0 (stearate/palmitate) profile of PC and PE; *"LPGAT1 knockout decreased the incorporation of stearate into PC and PE."*

Structural/database orientation. UniProt Q92604 curates the enzyme as an ER-membrane, PlsC/AGPAT-family acyltransferase (Pfam PF01553/PF16076; InterPro IPR002123) with the catalytic HX4D motif (residues ~101–106) and two transmembrane helices. The primary curated catalytic reaction is $2\text{-acyl-LPE} + \text{stearoyl-CoA} \rightarrow 1\text{-stearoyl-2-acyl-PE}$.

Together these establish the **core MF** (sn-1 LPE/LPC O-acyltransferase, stearoyl-CoA-preferring), **core BP** (PE/PC sn-1 acyl-chain remodeling), and **core CC** (ER membrane).

Finding 2 — The secondary activities (MGAT, PG, PI, PS, TAG) are real but weak, context-specific, or in-vivo-only

Activity	Best evidence	Nature	Curation status
MGAT / DAG synthesis	P 20018982 (mouse liver)	Direct assay, single group, mouse; not reproduced by sn-1 studies	UniProt reactions ECO:0000250 by-similarity to mouse Lpgat1; GO:0003846 ISS. Non-core.
TAG biosynthesis	P 20018982 ; P 42173283	Downstream pathway flux — KO <i>increases</i> TAG/lipid droplets	GO:0019432 ISS. Downstream phenotype, non-core.
LPG → PG remodeling	P 15485873 (human); P 37917582 (MEGDEL KO)	Experimentally real in human but LPG is a <i>poor</i> intrinsic substrate vs LPE/LPC	9 PG reactions ECO:0000269 P 15485873 . Minor, shared with paralogs.
LPI → PI remodeling	none direct; LPI "poor" (P 36049524); no activity in 2004 assay (P 15485873)	No curated UniProt reaction; ALCAT1 carries LPIAT (P 19075029)	GO:0036149 IBA only. Flag for removal/generalization.
1-acyl-LPS acylation	P 36049524 ("weak activities"); P 42173283 (in vivo PS reduction)	Weakly experimental + in-vivo lipidomic	RHEA:74815/37403 ECO:0000269 P 36049524 . Retain with weak/secondary qualifier.

MGAT/TAG. [PMID:20018982](#) identified mouse LPGAT1 as an acyl-CoA:monoacylglycerol acyltransferase; hepatic MGAT activity fell with LPGAT1 shRNA and serum TAG dropped ("*we identified a novel MGAT gene, which is identical with lysophosphatidylglycerol acyltransferase1 (LPGAT1)*"). This is direct but restricted to mouse liver, from a single group, and not reproduced

by the later sn-1-focused studies. Crucially, [PMID:42173283](#) shows that LPGAT1/Lplat7 knockout **increases** triglyceride and lipid-droplet accumulation — the *opposite* of what a direct, rate-limiting TAG synthase would produce on deletion. TAG "biosynthesis" is therefore best read as a **downstream pathway-flux/abundance phenotype**, not intrinsic chemistry.

PG remodeling. The original human enzyme characterization ([PMID:15485873](#)) demonstrated LPG acyltransferase activity, and knockout produces aberrant PG acyl composition and a MEGDEL-like disease ([PMID:37917582](#)). Yet LPG is a *poor* intrinsic substrate relative to LPE/LPC ([P 36049524](#)), and

[P 35131264](#) explicitly shows acylation into LPE "rather than" LPG. So PG remodeling is experimentally real in human but **minor in intrinsic ranking**, and is shared with ALCAT1/LCLAT1 and CRLS1.

PI remodeling. LPI is a poor substrate ([P 36049524](#)), and the 2004 human assay found *no* LPI activity: "*no significant acyltransferase activities were detected against glycerol 3-phosphate or a variety of lysophospholipids, including lysophosphatidylcholine, lysophosphatidylethanolamine, lysophosphatidylinositol, and lysophosphatidylserine.*" (Note this early assay also missed LPC/LPE, so absence there reflects assay limitations — but the PI negative is consistent with later "poor substrate" data.) The same-family paralog ALCAT1/LCLAT1 carries robust LPIAT activity ([PMID:19075029](#)), creating a **paralog-attribution risk** for the PI-remodeling annotation (GO:0036149, IBA only, no curated UniProt reaction).

LPS acylation. No isolated in vitro human assay demonstrates lyso-PS acylation as a strong activity; the RHEA-curated PS reactions are attributed to [P 36049524](#)'s "weak activities toward other lyso-PLs." In vivo support comes from

[P 42173283](#): Lplat7 knockout "*reduced the abundance of 1-stearoyl-2-unsaturated species of phosphatidylcholine, phosphatidylethanolamine, and phosphatidylserine*" — supporting an sn-1 PS remodeling contribution in vivo, alongside PC and PE.

Finding 3 — RHEA/UniProt evidence codes refine sn-position and provenance of the secondary activities

A programmatic pull of UniProt Q92604 catalytic-activity comments with RHEA IDs and ECO codes sharpens the picture in ways that matter directly for curation:

- **PS acylation is curated as EXPERIMENTAL** (ECO:0000269, PubMed:36049524), not merely electronic. Two reactions exist: **RHEA:74815** (sn-1 acylation of 2-acyl-LPS, adding palmitoyl) and **RHEA:37403** (sn-2 acylation of 1-oleoyl-LPS, adding stearoyl). The seed's exact phrase "*acylates 1-acyl-lysophosphatidylserine*" maps to **RHEA:37403 (sn-2 addition onto a 1-acyl acceptor)** — an important distinction from the enzyme's sn-1 preference on LPE/LPC.
- **MGAT/DAG reactions** (RHEA:16741/65096/65100, 2-acylglycerol → 1,2-diacylglycerol) are **ECO:0000250 by-similarity to MOUSE** Lpgat1 (UniProtKB:Q91YX5) — i.e., **no human experimental support** in UniProt.
- **All 9 PG reactions** are ECO:0000269 PubMed:15485873 (human experimental, 2004).
- **LPE/LPC sn-1 reactions** are ECO:0000269 PubMed:36049524.
- **UniProt curates NO lysophosphatidylinositol (LPI) reaction** for LPGAT1, despite GO:0036149 (PI remodeling) existing as an IBA annotation.

Paralog check. LPGAT1 (PANTHER PTHR10983:SF2) and LCLAT1/ALCAT1 (Q6UWP7, PTHR10983:SF16) are same-family AGPAT/PlsC paralogs sharing Pfam PF01553+PF16076 and InterPro IPR002123/IPR032098. CRLS1 (Q9UJA2) is a distinct family (PTHR14269, PF01066). This supports the interpretation that anionic-lysophospholipid (LPI/LPG) activities historically attributed to LPGAT1 may partly belong to ALCAT1.

Mechanistic Model / Interpretation

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                Lands cycle (sn-1 remodeling), ER membrane
                =====
2-acyl-LPE + stearoyl-CoA --LPGAT1--> 1-stearoyl-2-acyl-PE [CORE, experimental x3]
2-acyl-LPC + stearoyl-CoA --LPGAT1--> 1-stearoyl-2-acyl-PC [CORE, experimental]

Preference:  stearoyl-CoA (18:0) > palmitoyl-CoA (16:0)
Position:    sn-1 specific on LPE/LPC
Phenotype:   KO shifts PC/PE from 18:0 -> 16:0/18:1 (reciprocal)

                Secondary / re-ranked capacities
                =====
2-acyl-LPS --LPGAT1--> PS (RHEA:74815, sn-1) [WEAK experimental, <a href="https://pubmed.ncbi.nlm.nih.gov/42173283/" rel="noopener">https://pubmed.ncbi.nlm.nih.gov/42173283/ re
1-acyl-LPS --LPGAT1--> PS (RHEA:37403, sn-2) [<= seed's exact phrase; WEAK]
LPG         --LPGAT1--> PG (RHEA x9)          [human experimental but POOR substrate;
                                                shared with ALCAT1/CRLS1]
2-MAG       --LPGAT1--> DAG -> TAG            [MOUSE by-similarity only; TAG = flux]
LPI         --??-->      PI                    [NO curated reaction; likely ALCAT1]

Downstream / abundance phenotypes (NOT intrinsic chemistry):
- KO INCREASES TAG / lipid droplets (<a href="https://pubmed.ncbi.nlm.nih.gov/42173283/" rel="noopener">https://pubmed.ncbi.nlm.nih.gov/42173283/ re
- KO causes MEGDEL-like disease via PG remodeling + mito transport (<a href="https://pubmed.ncbi.nlm.nih.gov/42173283/" rel="noopener">https://pubmed.ncbi.nlm.nih.gov/42173283/ re

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The unifying model: **LPGAT1 is a Lands-cycle sn-1 acyl remodeler whose bona-fide, reproducible substrates are LPE and LPC with a stearoyl-CoA preference.** Its promiscuity toward anionic and other lysophospholipids (LPG, LPI, LPS) is genuine but intrinsically weak; whether a given weak activity produces a physiological phenotype depends on tissue context, substrate availability, and the presence of overlapping paralogs (ALCAT1, CRLS1). The dramatic in-vivo phenotypes (MEGDEL syndrome, TAG accumulation, reproductive defects) are **systems-level consequences of perturbing membrane acyl composition and mitochondrial lipid supply**, not direct evidence that PG/TAG/PS synthesis is the enzyme's primary chemistry.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
PMID:36049524	Direct assay + KO lipidomics	Supports core; qualifies secondary	sn-1 LPE/LPC is primary; other lyso-PLs weak	LPLAT7 mainly acylates sn-1 of LPC/LPE, weak elsewhere; only C18:0-PC/PE reduced in KO	Human recombinant + cells/tissue
PMID:35131264	Direct assay + KO	Supports core; refutes strong PG	sn-1 LPE specificity, stearyl>palmitoyl	Acyl-CoA into sn-1 LPE "rather than" LPG; KO shifts PE/PC 18:0→16:0	Murine recombinant (bacterial)
PMID:37217003	Mutant phenotype (in vivo)	Supports core BP	sn-1 18:0/16:0 remodeling of PC/PE	KO decreases stearate incorporation into PC/PE	Mouse skeletal muscle
PMID:20018982	Direct assay + shRNA	Qualifies MGAT/TAG	MGAT capacity, TAG secretion	LPGAT1 = novel MGAT; shRNA lowers hepatic MGAT + serum TAG	Mouse liver
PMID:15485873	Direct assay	Supports PG; refutes LPI/LPS (era-limited)	Original human LPGAT characterization	LPG activity present; NO LPI/LPS/G3P/LPC/LPE activity detected	Human recombinant, ER
PMID:37917582	Mutant phenotype	Qualifies PG (downstream)	PG remodeling → mitochondrial function	KO causes MEGDEL-like disease; PG ER-retention, crista loss	Mouse KO
PMID:42173283	KO lipidomics	Qualifies PS; refutes TAG-synthase	sn-1 remodeling of PC/PE/PS; TAG	KO reduces 1-stearyl-2-unsat PC/PE/PS; INCREASES TAG/droplets	Mouse KO
PMID:19075029		Competing (paralog)	Who carries LPIAT/LPGAT?	ALCAT1/LCLAT1 overexpression	Human HEK293

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
	Direct assay (paralog)			increases LPIAT + LPGAT activity	
PMID:20025994	Direct assay (paralog CLS1)	Competing (paralog)	PG remodeling attribution	hCLS1/CRLS1 has strong LPG acyltransferase activity	Human recombinant
UniProt Q92604 / RHEA	Database + ECO codes	Qualifies provenance	Reaction-level evidence codes	PS=experimental(269); MGAT=by-similarity mouse(250); PG=269; no LPI reaction	Curated database

GO Curation Implications

The following are **leads requiring curator verification**.

Retain as core: - **MF:** sn-1-specific lysophospholipid O-acyltransferase acting on LPE/LPC (e.g., 1-acylglycerophosphoethanolamine O-acyltransferase; stearyl-CoA preferring). Evidence:

[P 36049524](#)

[P 35131264](#)

(both ECO:0000269-grade). - **BP:** phosphatidylethanolamine/phosphatidylcholine acyl-chain remodeling (Lands cycle, sn-1). Evidence:

[P 37217003](#)

[P 35131264](#)

- **CC:** endoplasmic reticulum membrane. Evidence: UniProt Q92604,

[P 15485873](#)

Retain with a weak/secondary qualifier: - MF: 1-acylglycerophosphoserine O-acyltransferase (GO:0106263 / RHEA:37403,74815). Backed by experimental evidence (ECO:0000269,

P 36049524

but explicitly the "weak activities toward other lyso-PLs." Curator should note it is secondary and that the seed's "1-acyl-LPS" phrasing corresponds to the sn-2 reaction RHEA:37403.

Demote to non-core (retain but flag as by-similarity / context-specific): - MGAT (GO:0003846) and TAG biosynthesis (GO:0019432). MGAT reactions are ECO:0000250 by-similarity to mouse only; TAG "biosynthesis" is a downstream flux phenotype (KO *raises* TAG). Neither should be presented as a primary human function.

Flag for removal or generalization: - PI remodeling (GO:0036149, IBA only). No curated UniProt reaction; LPI is a poor substrate; ALCAT1/LCLAT1 is a strong LPIAT paralog. High risk of paralog carry-over. Recommend removal or generalization to "glycerophospholipid remodeling."

PG **remodeling:** retain (human experimental,

P 15485873

but annotate as minor/shared, given LPG is a poor substrate relative to LPE/LPC and the activity overlaps with ALCAT1/CRLS1.

Avoid "protein binding" as a final recommendation — the informative MF terms above are supported. Note: LPGAT1's interaction with the prohibitin/TIMM14 import motor (

P 37917582

could justify a specific protein-binding annotation with a named partner, but that is orthogonal to the acyltransferase function under review.

Mechanistic Scope

The immediate molecular function under test is **acyl-CoA-dependent O-acyltransferase chemistry on lysophospholipid acceptors**. The direct, gene-product-intrinsic activity is transfer of a (preferentially stearyl) acyl chain onto the sn-1 position of LPE and LPC. Everything else in the seed hypothesis must be separated from this:

- **Direct chemistry (intrinsic):** sn-1 LPE/LPC acylation (strong); LPG, LPI, LPS acylation (weak, "poor is not zero").
- **Pathway consequences / flux:** PG remodeling feeding cardiolipin synthesis and mitochondrial cristae; TAG abundance changes.
- **Downstream phenotypes / disease:** MEGDEL-like syndrome, dilated cardiomyopathy, 3-methylglutaconic aciduria

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P 37917582
);
 hepatic lipid accumulation

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);
 LUAD proliferation

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P 37258473
);
 zebrafish reproductive/developmental defects

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P 35508627
);
 BMI/body-fat associations

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P 23505186
);
 and plasma 18:0 GWAS signals

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P 23362303
).

Only the first category constitutes MF evidence. The second and third support BP and disease-association annotations but must not be curated as direct enzymatic function.

Conflicts and Alternatives

1. **Positional-specificity re-ranking vs exclusion.** The newer sn-1 LPE/LPC data

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do not exclude older secondary activities; they re-rank them. "Poor is not zero" is the correct reading. Construct (bacterial murine vs recombinant human vs endogenous), assay (in vitro vs cellular vs KO lipidomics), and organism differences explain most apparent contradictions.

2. **Paralog confusion (ALCAT1/LCLAT1; CRLS1).** ALCAT1 robustly carries LPIAT and LPGAT activities

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and is a same-family AGPAT/PlsC paralog; CRLS1/hCLS1 has strong LPG acyltransferase activity

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Anionic-lysophospholipid activities historically credited to LPGAT1 (especially PI and part of PG) may belong to these paralogs.

3. **Organism/tissue specificity.** MGAT/TAG evidence is mouse-liver

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);
sn-1 PC/PE evidence spans human recombinant, murine recombinant, and murine muscle. Human-experimental support for MGAT is limited.

4. **sn-position confound (acyl migration).** Lyso-phospholipid substrates undergo non-enzymatic acyl migration between sn-1 and sn-2, so "sn-1 vs sn-2" assignments (including the seed's "1-acyl-LPS" → RHEA:37403 sn-2 mapping) carry inherent uncertainty.

5. **Loss-of-function ≠ direct chemistry.** The MEGDEL-like disease

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TAG accumulation

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reproductive/developmental defects (zebrafish,

P 35508627

and cancer proliferation phenotypes

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are downstream/pleiotropic consequences, not evidence of primary MGAT/TAG/PG chemistry.

6. Era-limited negative results.

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found no LPC/LPE activity — later shown to be the core function — so its negative LPI/LPS results should be weighted as assay-limited, not definitive.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Human LPS acylation kinetics	RHEA reactions curated from P 36049524 "weak activities"	Determines whether GO:0106263 is a genuine human MF or trace promiscuity	Purified human LPGAT1 kinetic assay on defined 1-acyl- vs 2-acyl-LPS with acyl-migration controls
Human MGAT activity	Only P 20018982 (mouse liver); UniProt by-similarity	Determines if MGAT/TAG deserves any human MF annotation	Recombinant human LPGAT1 MGAT assay; independent replication
PI/LPI attribution	No curated reaction; ALCAT1 carries LPIAT	GO:0036149 may be paralog carry-over	Side-by-side LPGAT1 vs ALCAT1 LPIAT assay; PI lipidomics in LPGAT1-specific KO
Relative flux of PG vs PE/PC remodeling in vivo	KO phenotypes exist for both	Whether PG remodeling is quantitatively minor	Quantitative flux/isotope tracing across lipid classes in a single KO model
sn-position robustness	In vitro assays report sn-1/sn-2	Acyl migration undermines strict positional claims	Rapid-quench/MS with migration-locked substrate analogs

This report relied on the supplied knowledge outline (17 papers, 3 confirmed findings) and public database/RHEA provenance; no new wet-lab data were generated. sn-position claims inherit acyl-migration uncertainty. Several key phenotypic studies are murine or single-group. UniProt/RHEA evidence codes were used as provenance orientation and specific reaction IDs should be re-verified against the current UniProt release before curation.

Proposed Follow-up Experiments / Discriminating Tests

1. **Purified human LPGAT1 kinetic panel:** measure $V_{\max}/K_m$ for LPE, LPC, LPG, LPI, LPS (both 1-acyl and 2-acyl regioisomers) and 2-MAG with matched acyl-CoA donors, including acyl-migration controls, to rank intrinsic activities on a common scale.
2. **Paralog side-by-side:** express human LPGAT1 vs ALCAT1/LCLAT1 (and CRLS1) in the same system and assay LPIAT/LPGAT to resolve PI/PG attribution.
3. **Isoform/organism control:** compare human vs mouse enzyme MGAT activity under identical conditions to test whether MGAT is species-specific.
4. **In vivo flux vs abundance:** isotope-tracing lipidomics in an LPGAT1-specific KO to separate direct acylation flux from steady-state abundance changes (especially for TAG and PS).
5. **Structural docking / active-site modeling:** use the AGPAT/PlsC fold and HX4D motif to predict headgroup selectivity and rationalize weak anionic-lysophospholipid activity.

Curation Leads (require curator verification)

Candidate	references	+	exact	snippets	to	verify:	-
P 36049524							
— "mainly incorporated several fatty acids into the sn-1 position of lysophosphatidylcholine (LPC) and lysophosphatidylethanolamine (LPE), with weak activities toward other lyso-PLs" → supports core MF + "poor is not zero."							
P 35131264							
— "transferred saturated acyl-CoAs specifically into the sn-1 position of lysophosphatidylethanolamine (LPE) rather than lysophosphatidylglycerol and preferred stearoyl-CoA over palmitoyl-CoA" → supports sn-1 LPE core; refutes strong PG.							
P 20018982							
— "we identified a novel MGAT gene, which is identical with lysophosphatidylglycerol acyltransferase1 (LPGAT1)" → MGAT, but mouse liver / by-similarity in UniProt.							
P 15485873							
— "no significant acyltransferase activities were detected against ... lysophosphatidylinositol, and lysophosphatidylserine" → historical negative for LPI/LPS.							

P 42173283

— *"reduced the abundance of 1-stearoyl-2-unsaturated species of phosphatidylcholine, phosphatidylethanolamine, and phosphatidylserine"* → in-vivo PS support; note KO raises TAG. -

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— *"HEK293 cells overexpressing human ALCAT1 demonstrated significant increases in LPI acyltransferase (LPIAT) and LPG acyltransferase (LPGAT) activities"* → paralog alternative.

Candidate GO actions: - Keep: sn-1 LPE/LPC acyltransferase MF; PE/PC remodeling BP; ER membrane CC. - Keep with weak qualifier: 1-acylglycerophosphoserine O-acyltransferase (GO:0106263). - Demote to non-core: MGAT (GO:0003846), TAG biosynthesis (GO:0019432). - Flag for removal/generalization: PI remodeling (GO:0036149, IBA). - Retain but mark minor/shared: PG remodeling.

Suggested curator questions: - Is any human-experimental MGAT activity available beyond

P 20018982

- Should GO:0036149 (PI) be reassigned to ALCAT1/LCLAT1? - Does the RHEA:37403 (sn-2 on 1-acyl-LPS) mapping justify the seed's "acylates 1-acyl-lysophosphatidylserine" as a curated MF, given only "weak activity" evidence?

Suggested experiments: the kinetic panel and paralog side-by-side assays under Discriminating Tests.

Conclusion

The seed hypothesis is **partially supported**. LPGAT1's primary, reproducible function is sn-1-specific LPE/LPC acylation with stearoyl-CoA preference at the ER. The proposed secondary capacities are genuinely present in the literature but are **re-ranked, not erased**, by the newer positional-specificity data: LPS acylation is weakly experimental (retain with qualifier), MGAT/TAG are by-similarity/downstream (demote to non-core), PG remodeling is minor and shared, and PI remodeling is best attributed to the ALCAT1 paralog (flag for removal). Construct, assay, and organism differences — not mutual exclusion — explain the apparent tension between older and newer studies.

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